12. M_p denotes the mass of the proton and M_n denotes mass of a neutron. A given nucleus of binding energy B, contains Z protons and N neutrons. The mass M(N,Z) of the nucleus is given by(where c is the speed of light)

(a) $M(N,Z) = NM_n + ZM_p - Bc^2$

(b) $M(N,Z) = NM_n + ZM_p + Bc^2$

(c) $M(N,Z) = NM_n + ZM_p - B/c^2$

(d) $M(N,Z) = NM_n + ZM_p + B/c^2$

13. A radioactive nucleus (initial mass number A and atomic number Z) emits two α-particles and 2 positrons. The ratio of number of neutrons to that of proton in the final nucleus will be

(a) $\frac{A-Z-4}{Z-2}$ (b) $\frac{A-Z-2}{Z-6}$

(c) $\frac{A-Z-4}{Z-6}$ (d) $\frac{A-Z-12}{Z-4}$

14. The half-life period of a radioactive element A is same as the mean life time of another radioactive element B. Initially both have the same number of atoms. Then

(a) A and B have the same decay rate initially

(b) A and B decay at the same rate always

(c) B will decay at faster rate than A

(d) A will decay at faster rate than B.

15. A radiative element has N_0 number of nuclei at t=0. The number of nuclei remaining after half of a half-life (that is, at time $t = \frac{1}{2}T_{\frac{1}{2}}$)

(a) $\frac{N_0}{2}$ (b) $\frac{N_0}{\sqrt{2}}$

(c) $\frac{N_0}{4}$ (d) $\frac{N_0}{8}$

Answers

1) b	**2)** d	**3)** c	**4)** c	**5)** b
6) b	**7)** d	**8)** b	**9)** c	**10)** A
11) b	**12)** c	**13)** b	**14)** c	**15)** b

II Short answer questions

1. What are cathode rays?.

2. Write the properties of cathode rays.

3. Give the results of Rutherford alpha scattering experiment.

4. Write down the postulates of Bohr atom model.

5. What is meant by excitation energy.

6. Define the ionization energy and ionization potential.

7. Write down the draw backs of Bohr atom model.

8. What is distance of closest approach?

9. Define impact parameter.

10. Write a general notation of nucleus of element X. What does each term denote?

11. What is isotope? Give an example.

12. What is isotone? Give an example.

13. What is isobar? Give an example.

14. Define atomic mass unit *u*.

15. Show that nuclear density is almost constant for nuclei with $Z > 10$.

16. What is mass defect?

17. What is binding energy of a nucleus? Give its expression.

18. Calculate the energy equivalent of 1 atomic mass unit.

19. Give the physical meaning of binding energy per nucleon.

20. What is meant by radioactivity?

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